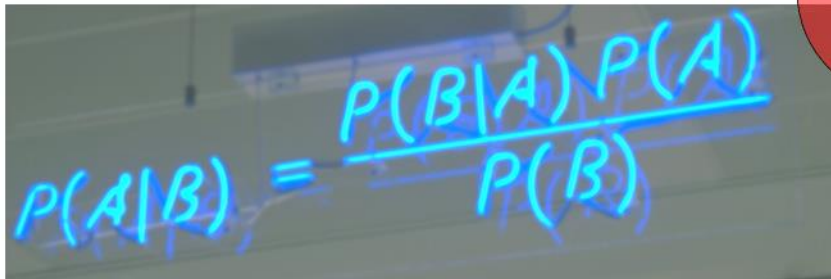


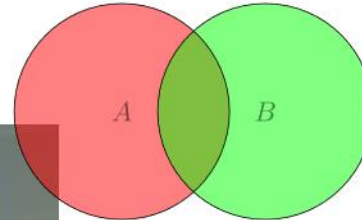
Probability: Part III

Conditional Probability & Bayes Theorem



A photograph of a whiteboard with the formula for conditional probability written in blue marker. The formula is $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$. The text is slightly blurry and has some blue lines drawn through it.

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$



Conditional Probability

- Probability measures how likely an event is to happen. But in the real world, events rarely happen in a vacuum. *Conditional probability* is the probability of an event occurring *given that another event has already occurred*.
- When you gain new information, your uncertainty changes. Conditional probability is the mathematical way of updating your beliefs based on new evidence.
- Imagine rolling a standard six-sided die.
- **Standard Probability:** What is the probability of rolling a 2? There is one "2" on a six-sided die, so the probability is $\frac{1}{6}$.
- **Conditional Probability:** Let's say I roll the die, cover it with my hand, and tell you: "*The number is even.*" What is the probability that the number is a 2?
- Because you *know* the number is even (Event B), your universe of possible outcomes just shrank. The only possible outcomes are now 2, 4, or 6.
- Since 2 is one of those three remaining possibilities, your new conditional probability is $\frac{1}{3}$.



Conditional Probability

- Real life application:
- Imagine Spotify wants to guess the probability that you will like a newly released track.
- *Unconditional Probability*: Without knowing anything about you, there is only a modest chance you will enjoy a completely random song.
- *Conditional Probability*: The moment Spotify adds the condition, *given* your listening history, the "universe" of possible songs *shrinks* drastically to only songs that fit your taste profile.
- Kew Question: How should probabilities be adjusted to consider this information?

Conditional Probability: Definition

- **Conditional probability** is the likelihood of an event occurring, *given* that another event has already happened.
- Essentially, knowing that one event has occurred gives you extra information, which shrinks your "universe" of possible outcomes and changes the odds of the next event.

The Formula

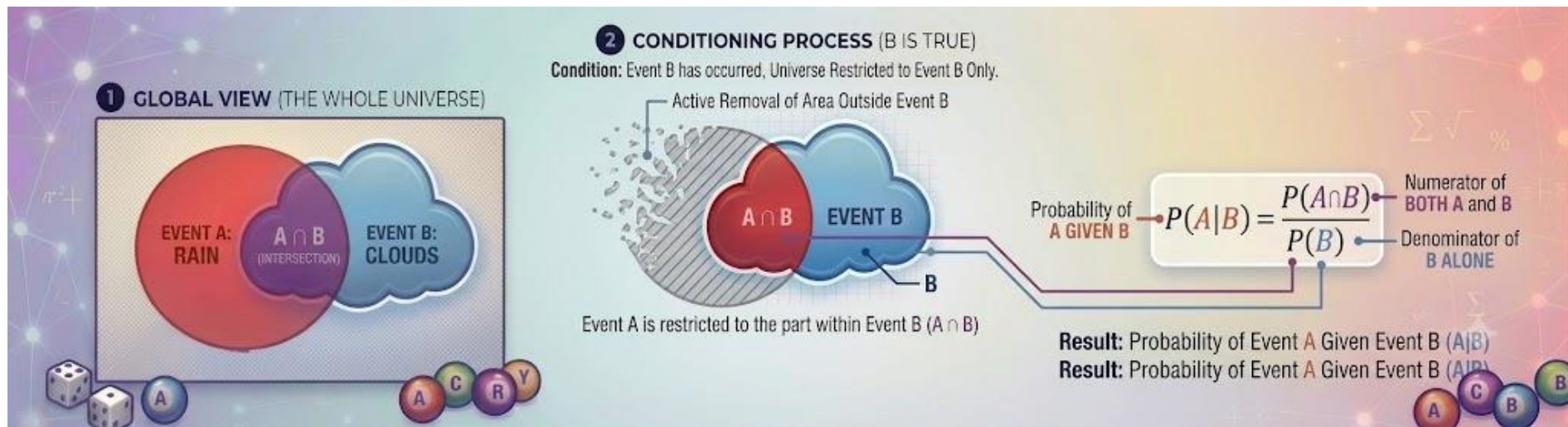
In mathematical terms, the conditional probability of Event A happening given that Event B has already happened is written as $P(A | B)$. It is calculated using this formula:

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

Note: We must assume that $P(B) \neq 0$

Conditional Probability: Intuition

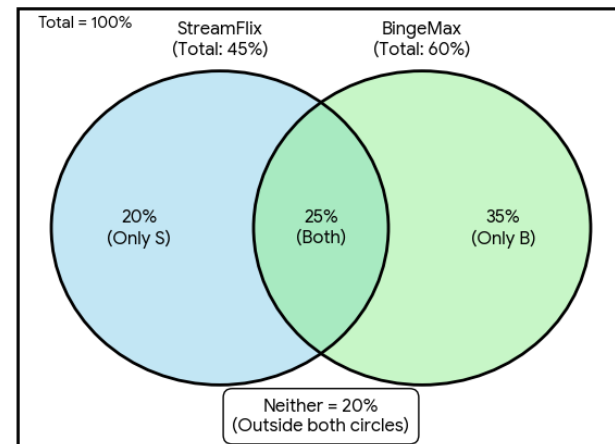
- If we're told that event B has occurred, then the sample space is restricted to B .
- The probability within B must be normalized, This is achieved by dividing by $P(B)$
- The event A can now only occur if the outcome is in of $A \cap B$.
Hence the new probability of A is: $P(A | B) = \frac{P(A \cap B)}{P(B)}$



Example 1

- A marketing firm is analyzing the subscription habits of households for two major streaming platforms: StreamFlix and BingeMax.
- The probability that a household subscribes to StreamFlix is 45%, the probability they subscribe to BingeMax is 60%, and the probability they subscribe to both is 25%.
- If it is known that a randomly selected household subscribes to **at most one** of these platforms, what is the probability that they subscribe to **neither**?

Streaming Subscriptions Venn Diagram



Independence: Definition

- Two events, A and B , are formally defined as independent if their joint probability (the chance of both happening together) is exactly equal to the product of their individual probabilities:

$$P(A \cap B) = P(A) \cdot P(B)$$

- This concept is often easier to understand through **conditional probability**. If A and B are independent, knowing that B has already happened doesn't change the odds of A happening:

$$P(A|B) = P(A)$$

$$P(B|A) = P(B)$$

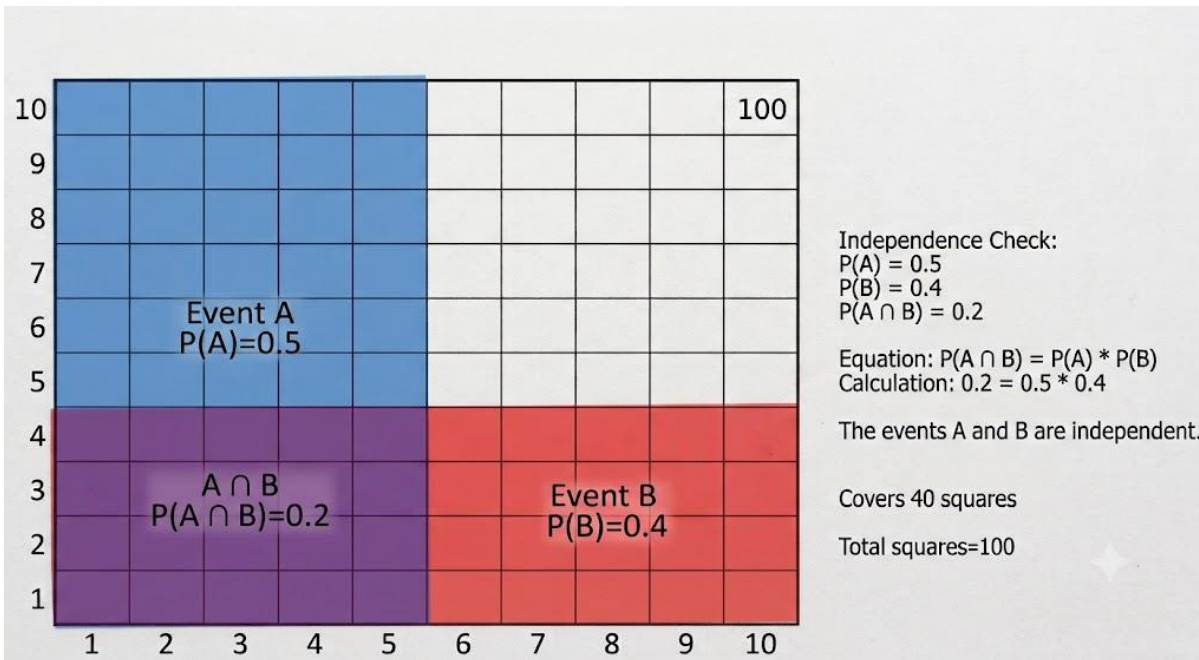
- (Note: This assumes that neither $P(A)$ nor $P(B)$ is zero).*

Thus, in the case of independence the conditional probability of an event is not affected by the knowledge of the other event

Independent Events Visualization

From the definition of independence, we have:

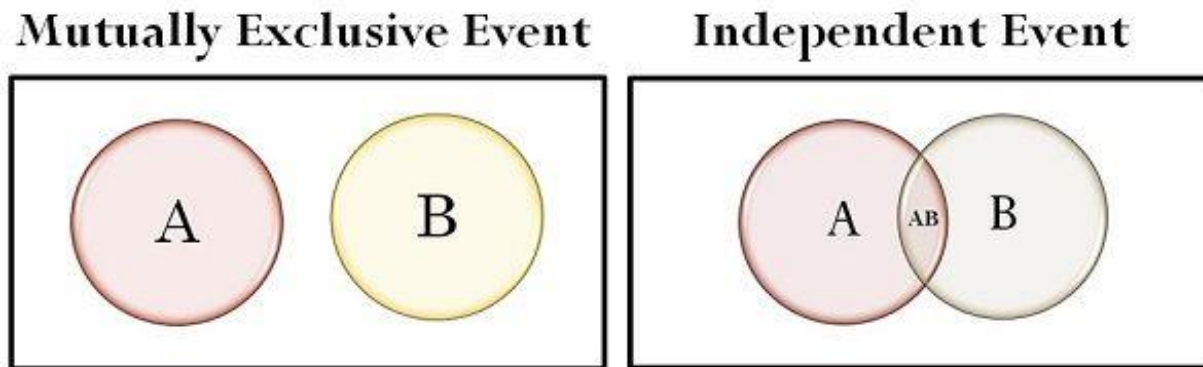
$$P(A \cap B) = P(A)P(B)$$
$$\Leftrightarrow \frac{P(A \cap B)}{P(B)} = \frac{P(A)}{1} = \frac{P(A)}{P(S)}$$



The ratio of the probability of the event A within B is the same as the ratio of the probability of the event A within the entire sample S .

Connection between independent and mutually exclusive events

Mutually exclusive events can *not* be independent unless the special case that at least one of them has probability zero.



In general, mutually exclusive events are highly *dependent* otherwise. A and B **cannot** occur simultaneously. If one event occurs the other event does not occur.

Notes

Definition extends to more than two events: A collection of n events $A_1, A_2, \dots, A_n$ is **mutually independent** if, for every sub-collection of k events (where $2 \leq k \leq n$) indexed by indices $\{i_1, i_2, \dots, i_k\}$, the probability of their intersection equals the product of their individual probabilities:

$$P\left(\bigcap_{j=1}^k A_{i_j}\right) = \prod_{j=1}^k P(A_{i_j})$$

Types of typical questions

- Check if two (or more) events A and B are independent:

Step 1: Compute $P(A)$, $P(B)$ and $P(A \cap B)$

Step 2: Check if

$$P(A \cap B) = P(A) * P(B)$$

- *Alternatively:* You may know the events are independent from the context of the problem and then use the above formula.

Example 2

Let A , B , and C be mutually independent, with $P(A) = P(B) = P(C) = 0.1$. Compute

a) $P(A \cup B)$

b) $P(A \cup B \cup C)$

Exercise:

c) Find the probability that exactly one of the events A , B , C will occur.

Example 3

A fair coin is tossed four times.

$$S = \{ HHHH, HHHT, HHTH, HHTT, HTHH, HTHT, HTTH, HTTT, \\ THHH, THHT, THTH, THTT, TTHH, TTHT, TTTH, TTTT \}$$

Each outcome is equally likely (prob = $1/16$). Let's define three distinct events:

Event A : The first two tosses are the same (either HH or TT).

Event B : There is an odd number of Heads in total (exactly 1 or 3 Heads).

Event C : The last two tosses are the same (either HH or TT).

Question: Are events A , B , and C mutually independent?

Exercise 1:

Suppose identical tags are placed on both ears of a fox. The fox is then let loose for a period of time. Consider the events

$C_1 = \{\text{left ear tag is lost}\}$ and $C_2 = \{\text{right ear tag is lost}\}$.

Assume C_1 and C_2 are *independent* events with

$P(C_1) = P(C_2) = p$.

Find the probability that *exactly* one tag is lost *given at most* one tag is lost.

The multiplicative rule of probability

$$P[A \cap B] = \begin{cases} P[A]P[B|A] & \text{if } P[A] \neq 0 \\ P[B]P[A|B] & \text{if } P[B] \neq 0 \end{cases}$$

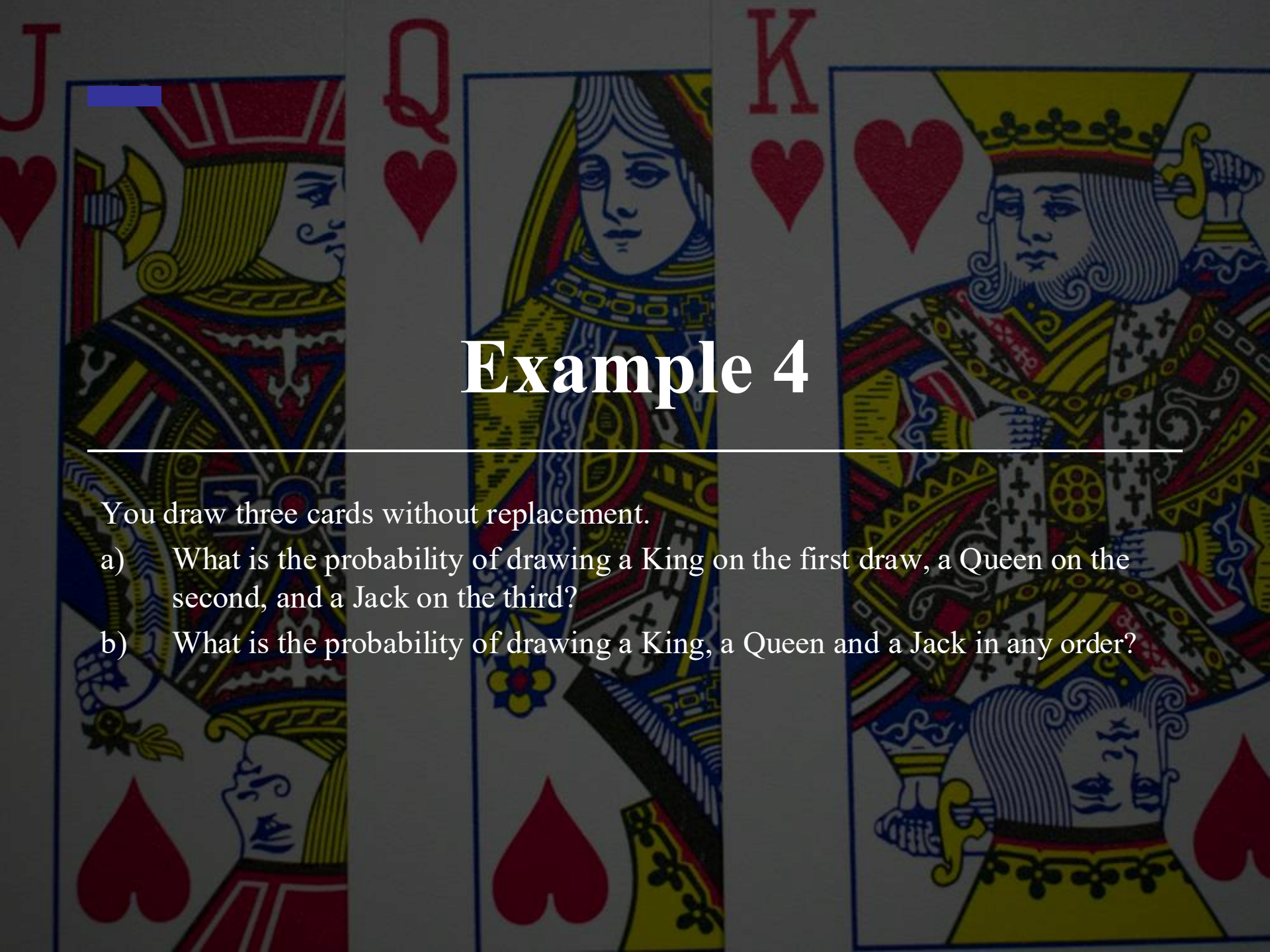
And simplifies to

$$P[A \cap B] = P[A]P[B]$$

only when A and B are **independent**.

Can be extended to more than two events:

$$\begin{aligned} & P(A_1 \cap A_2 \cap \cdots \cap A_n) \\ &= P(A_1) \times P(A_2|A_1) \times P(A_3|A_1 \cap A_2) \times \cdots \times P(A_n|A_1 \cap \cdots \cap A_{n-1}) \end{aligned}$$



Example 4

You draw three cards without replacement.

- a) What is the probability of drawing a King on the first draw, a Queen on the second, and a Jack on the third?
- b) What is the probability of drawing a King, a Queen and a Jack in any order?



Example 5: Urn model

An urn contains r red balls and b black balls.

A ball is selected at random, its color is noted, and it is *not* replaced.

A second ball is selected, and its color is noted.

- a) What is the probability that you first get a black ball and then a red ball?
- b) What is the probability that a red ball is selected on the second draw?

Example 5: Solution

a) Let:

- B_1 = the event that the first ball is black
- R_2 = the event that the second ball is red

Find $P(B_1 \cap R_2)$, the probability that the first ball is black, **and** the second ball is red.

Step 1: Apply the Multiplication Rule

Using conditional probability, the probability of both events occurring is:

$$P(B_1 \cap R_2) = P(B_1) \cdot P(R_2 \mid B_1)$$

Example 5: Solution

Step 2: Calculate $P(B_1)$

Initially, the urn contains b black balls out of a total of $r + b$ balls:

$$P(B_1) = \frac{b}{r + b}$$

Step 3: Calculate $P(R_2 \mid B_1)$

Given that B_1 has occurred, one black ball was removed and **not** replaced. The urn now contains r red balls out of a remaining total of $r + b - 1$ balls:

$$P(R_2 \mid B_1) = \frac{r}{r + b - 1}$$

Step 4: Compute the Final Result

$$P(B_1 \cap R_2) = \left(\frac{b}{r + b} \right) \cdot \left(\frac{r}{r + b - 1} \right) = \frac{br}{(r + b)(r + b - 1)}$$

b) To find $P(R_2)$, we note that the second ball can be red in two mutually exclusive cases:

The first ball was **red** AND the second ball is **red** ($R_1 \cap R_2$)

The first ball was **black** AND the second ball is **red** ($B_1 \cap R_2$)

$$P(R_2) = P(R_1 \cap R_2) + P(B_1 \cap R_2)$$

Step 1: Case 1 — First is Red, Second is Red ($R_1 \cap R_2$)

$$\begin{aligned} P(R_1 \cap R_2) &= P(R_1) \cdot P(R_2 | R_1) = P(R_1) = \left(\frac{r}{r+b}\right) \left(\frac{r-1}{r+b-1}\right) \\ &= \frac{r(r-1)}{(r+b)(r+b-1)} \end{aligned}$$

Step 2: Case 2 — First is Black, Second is Red ($B_1 \cap R_2$)

From the previous part, we know:

$$P(B_1 \cap R_2) = P(B_1) \cdot P(R_2 | B_1) = \frac{br}{(r+b)(r+b-1)}$$

Step 3: Combine and Simplify

Summing both cases together:

$$\begin{aligned} P(R_2) &= \frac{r(r-1)}{(r+b)(r+b-1)} + \frac{br}{(r+b)(r+b-1)} \\ &= \frac{r[(r-1)+b]}{(r+b)(r+b-1)} = \frac{r(r+b-1)}{(r+b)(r+b-1)} \end{aligned}$$

Cancel the common factor $(r+b-1)$ in the numerator and denominator:

$$P(R_2) = \frac{r}{r+b}$$

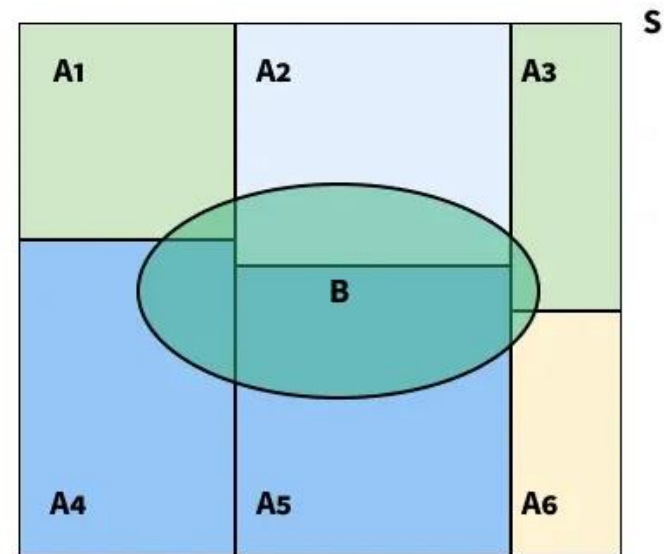
Example 5: Time to think

- If two balls are drawn sequentially without replacement, what is the probability that exactly one of the two balls is red?
- Suppose you draw every single ball out of the urn one by one without replacement until the urn is empty. What is the probability that the *very last* ball you draw is red?
- The technique used to solve part b) has a general name. It is called the *law of total probability*.

Rule: Law of Total Probability

- What we used above can be generalized to more than two events.
- It is known as the law of total probability
- Let $A_1, \dots, A_k$ be mutually exclusive and exhaustive events. That is, their union is the sample space S and none of them overlap (think partitioning S).
- Then for any event B

$$P(B) = P(A_1)P(B|A_1) + \dots + P(A_k)P(B|A_k)$$



Example 6 (World Cup)

For the Spain Men's National Soccer Team, comprehensive match-state metrics show they win roughly **90%** of the games in which they score first, reflecting their elite possession dominance and defensive control. Exact statistical tracking for when an opponent scores first is rare due to their tendency to secure clean sheets, though general international data sets project win rates dropping significantly to **20%** when conceding first. During recent competitive runs—including their triumphant FIFA World Cup campaign—the Spain national football team scored first in approximately **86%** of their matches, reflecting a dominant style of play that heavily relies on early possession and opening goals. Assume no ties.

- a) Find the probability the team wins a game.
- b) If they won what is the probability they scored first?

Bayes Rule

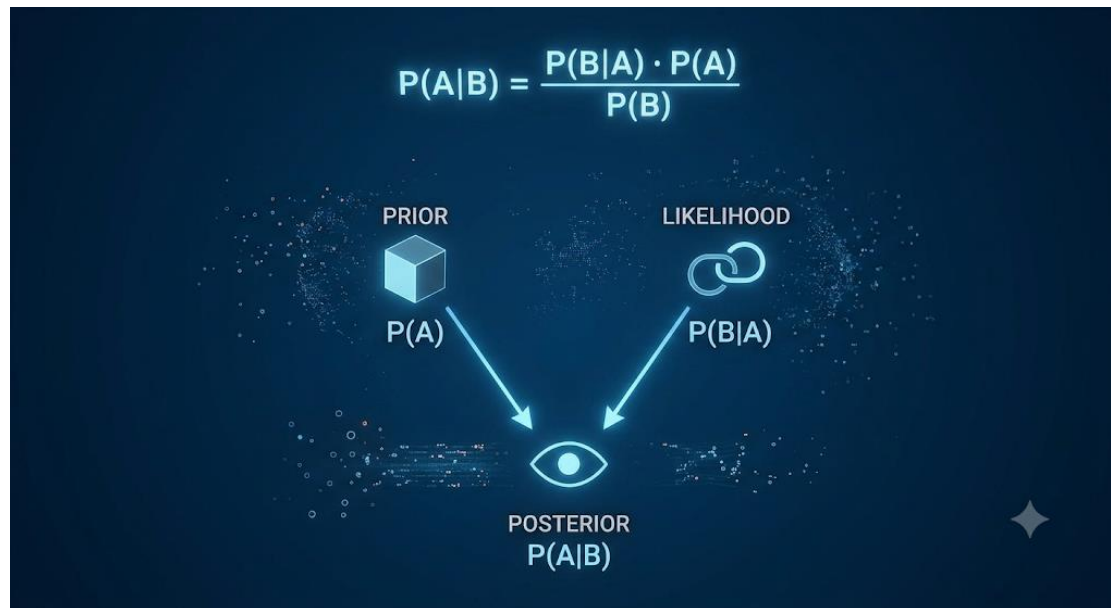
- If we assume that events A_1 and A_2 are mutually exclusive and exhaustive (so A_2 is the exact complement of A_1), to find the probability of event A_1 given event B , the Bayes' Rule is:

$$P(A_1 | B) = \frac{P(B | A_1)P(A_1)}{P(B | A_1)P(A_1) + P(B | A_2)P(A_2)}$$

- Similarly, if you want to find the probability of A_2 given B :

$$P(A_2 | B) = \frac{P(B | A_2)P(A_2)}{P(B | A_1)P(A_1) + P(B | A_2)P(A_2)}$$

- The denominator represents the Law of Total Probability



Bayes Rule: Proof

Step 1: The Definition of Conditional Probability

We begin with the standard definition of the conditional probability of A_1 given B :

$$P(A_1 \mid B) = \frac{P(A_1 \cap B)}{P(B)}$$

Bayes Rule: Proof

Step 2: Expanding the Numerator (The Multiplication Rule)

$$P(A_1 \cap B) = P(B \mid A_1)P(A_1)$$

Substituting this back into the numerator of our equation from Step 1 gives us the simple form of Bayes' Theorem:

$$P(A_1 \mid B) = \frac{P(B \mid A_1)P(A_1)}{P(B)}$$

Bayes Rule: Proof

Step 3: Partitioning the Denominator

Law of Total Probability for our specific two-event partition:

$$P(B) = P(B | A_1)P(A_1) + P(B | A_2)P(A_2)$$

Step 4: The Final Substitution

Finally, we substitute this expanded, two-part expression for $P(B)$ directly into the denominator of the equation we established at the end of Step 2:

$$P(A_1 | B) = \frac{P(B | A_1)P(A_1)}{P(B | A_1)P(A_1) + P(B | A_2)P(A_2)}$$

Example 7: Testing for a disease

Suppose that 0.1% of the population have a certain genetic disease.

A test is available to detect the disease.

If a person has the disease, the test concludes that he has the disease 96% of the time. If the person doesn't have the disease the test states that he has the disease 2% of the time (false positive).

Two properties of a medical test

Sensitivity = $P(\text{test is positive} \mid \text{disease}) = 0.96$

Specificity = $P(\text{test is negative} \mid \text{no disease}) = 1 - 0.02 = 0.98$

A person takes the test, and the test is positive, what is the probability that he (or she) has the disease?

Solution:

Let A = the event that the person has the disease

$\bar{A}$ = the event that the person doesn't have the disease

$$P[A] = 0.001, \quad P[\bar{A}] = 1 - 0.001 = 0.999$$

Let B = the event that the test is positive.

$$P[B|A] = 0.96, \quad P[B|\bar{A}] = 0.02 \quad \text{We want } P[A|B].$$

Note: Again, the **desired** conditional probability is in the reverse direction of the **given** conditional probabilities.

So, from Bayes Theorem

$$\begin{aligned} P(A|B) &= \frac{P(A \cap B)}{P(A)P(B|A) + P(\bar{A})P(B|\bar{A})} \\ &= \frac{0.001 \times 0.95}{0.001 \times 0.95 + 0.999 \times 0.02} = 0.045 \end{aligned}$$

The disease prevalence ($p = 0.001$) can be thought of as a ‘*prior*’ probability on the two outcomes:

Prior: $P(\text{disease}) = 0.001$ & $P(\text{no disease}) = 1 - 0.001 = 0.999$

Observing a positive test result causes us to modify this probability to $p = 0.045$ which is our ‘*posterior*’ probability that the patient has the disease:

Posterior: $P(\text{disease} | T+) = 0.045$ & $P(\text{no disease} | T+) = 0.955$

General Bayes Theorem

For a set of mutually exclusive and exhaustive events $A_1, A_2, \dots$ (i.e. $P(\cup_i A_i) = \sum_i P(A_i) = 1$), we have:

$$P(A_i|B) = \frac{P(B|A_i)P(A_i)}{\sum_{j=1}^{\infty} P(B|A_j)P(A_j)}$$

In Bayesian statistic the “parameter” is the indicator which one of the sets $A_1, A_2, \dots$ occurs with prior probabilities $P(A_i)$, which are then updated to posterior probabilities $P(A_i|B)$, after new “data” is observed from event B .



We have two urns:

- **Urn A** contains **12 red balls** and **8 black balls** (20 total).
- **Urn B** contains **4 red balls** and **16 black balls** (20 total).

An urn is selected at random with an equal 50% chance. From this chosen urn, **two balls are drawn sequentially without replacement**, and both turn out to be **red**.

Example

- **Part (a):** What is the posterior probability that the chosen urn was **Urn A** given that the first two balls drawn were both red?
- **Part (b):** If a **third ball** is now drawn from the *same* urn (without replacing the first two), what is the probability that this third ball will also be **red**?

Solution:

1. Define the Events

Let A be the event that Urn A was chosen: $P(A) = \frac{1}{2}$

Let B be the event that Urn B was chosen: $P(B) = \frac{1}{2}$

Let R_1R_2 be the event that the first two balls drawn are both red.

2. Calculate Conditional Likelihoods (Sampling Without Replacement)

If Urn A was chosen:

$$P(R_1R_2 \mid A) = \frac{12}{20} \times \frac{11}{19} = \frac{132}{380}$$

If Urn B was chosen:

$$P(R_1R_2 \mid B) = \frac{4}{20} \times \frac{3}{19} = \frac{12}{380}$$

3. Apply the Law of Total Probability to find $P(R_1R_2)$

$$P(R_1R_2) = P(R_1R_2 \mid A)P(A) + P(R_1R_2 \mid B)P(B)$$

$$P(R_1R_2) = \left(\frac{132}{380}\right)\left(\frac{1}{2}\right) + \left(\frac{12}{380}\right)\left(\frac{1}{2}\right) = \frac{144}{760} = \frac{18}{95}$$

4. Apply Bayes' Theorem

$$P(A \mid R_1R_2) = \frac{P(R_1R_2 \mid A)P(A)}{P(R_1R_2)} = \frac{\frac{132}{760}}{\frac{144}{760}} = \frac{132}{144} = \frac{\mathbf{11}}{\mathbf{12}}$$

$$\approx 91.67\%$$

$$(Correspondingly, P(B \mid R_1R_2) = 1 - \frac{11}{12} = \frac{1}{12} \approx 8.33\%)$$

Solution:

Part (b): Probability that the Third Ball is Red

Now we want to find $P(R_3 \mid R_1 R_2)$, which depends on which urn we actually have after updating our beliefs in Part (a).

1. Determine Remaining Contents in Each Urn

If Urn A (Probability $\frac{11}{12}$): 2 red balls were removed, leaving **10 red** and **8 black** (18 balls total).

$$P(R_3 \mid A, R_1 R_2) = \frac{10}{18}$$

If Urn B (Probability $\frac{1}{12}$): 2 red balls were removed, leaving **2 red** and **16 black** (18 balls total).

$$P(R_3 \mid B, R_1 R_2) = \frac{2}{18}$$

Solution:

2. Calculate the Total Predictive Probability

$$P(R_3 \mid R_1 R_2) \\ = P(R_3 \mid A, R_1 R_2) \cdot P(A \mid R_1 R_2) + P(R_3 \mid B, R_1 R_2) \cdot P(B \mid R_1 R_2)$$

$$P(R_3 \mid R_1 R_2) = \left(\frac{10}{18}\right)\left(\frac{11}{12}\right) + \left(\frac{2}{18}\right)\left(\frac{1}{12}\right)$$

$$P(R_3 \mid R_1 R_2) = \frac{110}{216} + \frac{2}{216} = \frac{112}{216} = \frac{14}{27} \approx 51.85\%$$

Exercise 2

A bank uses an automated AI system to monitor credit card transactions for fraud. Based on historical data, 95% of all transactions are legitimate and 5% are fraudulent. When a transaction is truly fraudulent, the AI correctly flags it as "suspicious" 85% of the time (and misses it, marking it "safe", 15% of the time). When a transaction is legitimate, the AI correctly marks it as "safe" 92% of the time (and incorrectly flags it as "suspicious" 8% of the time). Suppose a new transaction has just been flagged as "suspicious" by the system.

a) What is the probability that the transaction is legitimate?

b) To reduce the number of false alarms, the bank introduces a secondary, independent verification step. If the AI system flags a transaction as "suspicious", an automated SMS text is immediately sent to the cardholder asking them to verify the purchase.

If the transaction is truly legitimate, the cardholder successfully confirms it 90% of the time (and misses the text or fails to confirm 10% of the time).

If the transaction is truly fraudulent, the fraudster cannot confirm it 95% of the time (but manages to spoof a confirmation 5% of the time).

Suppose a new transaction is flagged as "suspicious" by the AI system AND the subsequent SMS text goes **unconfirmed**. Assuming the SMS confirmation behavior is independent of the AI system's flagging mechanism given the true nature of the transaction, what is the new probability that this transaction is **truly fraudulent**?



Example: Gambler's ruin

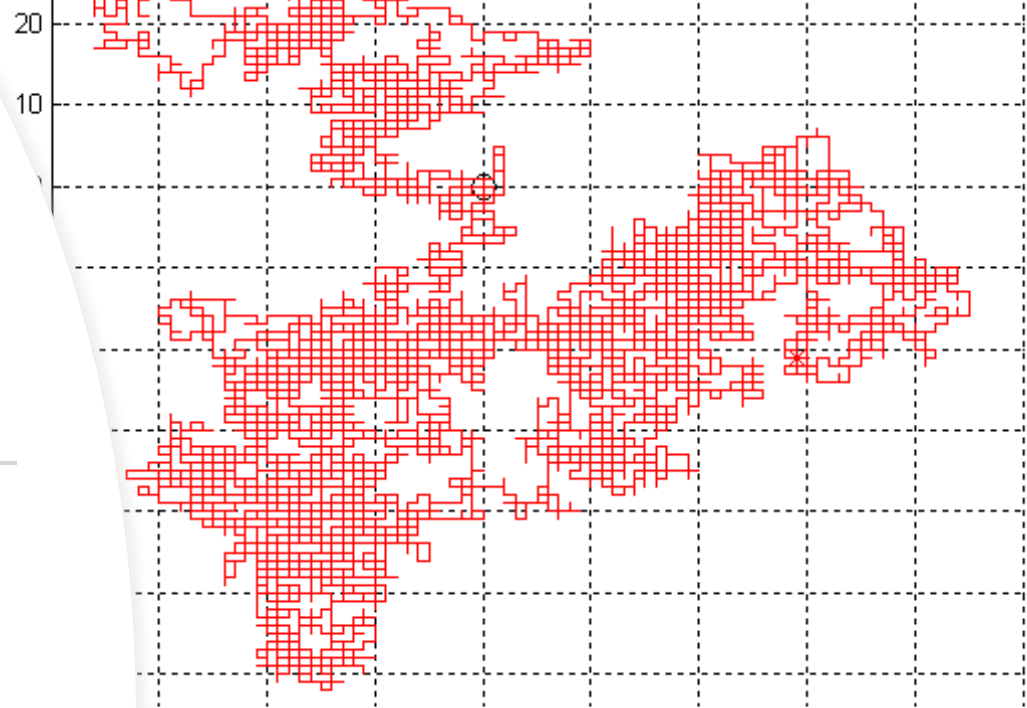
Suppose that two gamblers A and B are playing a game against each other. Let p be a given number ($0 < p < 1$), and suppose that on each play of the game, the probability that gambler A will win one dollar from gambler B is p and the probability that gambler B will win one dollar from gambler A is $1 - p$. Suppose also that the initial fortune of gambler A is i dollars, and the initial fortune of gambler B is $k - i$ dollars, where i and $k - i$ are given positive integers. Finally, suppose that the gamblers play the game repeatedly and independently until the fortune of one of them has been reduced to 0 dollars. Find the probability A wins.

Aside: Random Walk

If two statisticians were to lose each other in an infinite forest, the first thing they would do is get drunk. That way, they would walk randomly, which would give them the best chance of finding each other.

- **Pólya's Recurrence Theorem:** This is where the actual statistics come in. In 1921, mathematician George Pólya proved a theorem about random walks on a grid (like a giant piece of graph paper). He found that dimensionality matters immensely: **In 1D or 2D space (like walking on the flat surface of a forest):** A person doing a random walk will eventually visit *every single possible point* on the grid if given infinite time. The probability of returning to a starting point—or of two random walkers eventually crossing paths—is exactly 1 (a 100% guarantee).

Because the statisticians are trapped in a two-dimensional forest (moving N, S, E, or W), executing a random walk mathematically guarantees that their paths will eventually cross.



Example: Gambler's ruin

1. Setup and Boundary Conditions

- Let $q = 1 - p$ be the probability that gambler B wins a play.
- Let P_i denote the probability that gambler A ultimately wins the entire game (reaches k dollars) given that A currently has i dollars.
- The total amount of money in play is k dollars. The game ends when either A reaches k dollars (and B is ruined with 0) or A drops to 0 dollars. This gives us two boundary conditions:
 - $P_0 = 0$ (if A starts with 0 dollars, A has already lost)
 - $P_k = 1$ (if A reaches k dollars, A has won)

Example: Gambler's ruin

2. Formulating the Difference Equation

- For any fortune i such that $1 \leq i \leq k - 1$, we condition on the outcome of the very next play:
- With probability p , A wins 1 dollar and now has $i + 1$ dollars.
- With probability $q = 1 - p$, A loses 1 dollar and now has $i - 1$ dollars.
- Using the Law of Total Probability:

$$P_i = pP_{i+1} + qP_{i-1}$$

- Since $p + q = 1$, we write $P_i = (p + q)P_i$:

$$(p + q)P_i = pP_{i+1} + qP_{i-1}$$

- Rearranging terms:

$$p(P_{i+1} - P_i) = q(P_i - P_{i-1})$$

$$P_{i+1} - P_i = \frac{q}{p}(P_i - P_{i-1})$$

Example Solution: Gambler's ruin

3. Solving the Recurrence Relation

Notice that the differences $P_j - P_{j-1}$ form a geometric series with common ratio $\frac{q}{p}$.

Let's express each step in terms of P_1 :

$$P_2 - P_1 = \left(\frac{q}{p}\right) (P_1 - P_0) = \left(\frac{q}{p}\right) P_1$$

$$P_3 - P_2 = \left(\frac{q}{p}\right) (P_2 - P_1) = \left(\frac{q}{p}\right)^2 P_1$$

$\vdots$

$$P_i - P_{i-1} = \left(\frac{q}{p}\right)^{i-1} P_1$$

Summing these differences from $j = 1$ to i gives a telescoping sum:

$$P_i - P_0 = \sum_{j=1}^i (P_j - P_{j-1}) = P_1 \sum_{j=0}^{i-1} \left(\frac{q}{p}\right)^j$$

Since $P_0 = 0$:

$$P_i = P_1 \sum_{j=0}^{i-1} \left(\frac{q}{p}\right)^j$$

The fine print: Alternative solution of recurrence

- From the problem, we know the probability of winning from state i relies on the probabilities of winning from the adjacent states. Since p is the probability of winning a round and $q = 1 - p$ is the probability of losing, we have:

$$\begin{aligned}P_i &= pP_{i+1} + qP_{i-1} \\ pP_{i+1} - P_i + qP_{i-1} &= 0\end{aligned}$$

- We assume the solution has the form $P_i = r^i$ for some constant r . Substituting r^i into our difference equation gives:

$$pr^{i+1} - r^i + qr^{i-1} = 0$$

- Assuming $r \neq 0$, we can divide the entire equation by r^{i-1} to get a standard quadratic equation, which is our **characteristic equation**:

$$pr^2 - r + q = 0$$

- We can solve for r by factoring. Since $p + q = 1$, we can substitute 1 with $(p + q)$ in the middle term:

$$\begin{aligned}pr^2 - (p + q)r + q &= 0 \\ (pr - q)(r - 1) &= 0\end{aligned}$$

- This gives us two roots for r :

$$r_1 = 1, r_2 = \frac{q}{p}$$

- If $p \neq 0.5$, then $q \neq p$, which means our two roots $r_1 = 1$ and $r_2 = \frac{q}{p}$ are distinct. The general solution for a linear difference equation with distinct roots is:

$$\begin{aligned}P_i &= A(r_1)^i + B(r_2)^i \\ P_i &= A(1)^i + B\left(\frac{q}{p}\right)^i = A + B\left(\frac{q}{p}\right)^i\end{aligned}$$

Example: Gambler's ruin

4. Case Analysis

Case 1: Unfair Game ($p \neq \frac{1}{2}$, so $p \neq q$)

- Using the finite geometric series formula $\sum_{j=0}^{i-1} x^j = \frac{1-x^i}{1-x}$ with $x = \frac{q}{p}$

$$P_i = P_1 \frac{1 - (q/p)^i}{1 - q/p}$$

- To find P_1 , apply the boundary condition $P_k = 1$:

$$1 = P_1 \frac{1 - (q/p)^k}{1 - q/p} \Rightarrow P_1 = \frac{1 - q/p}{1 - (q/p)^k}$$

- Substituting P_1 back into the formula for P_i :

$$P_i = \frac{1 - \left(\frac{q}{p}\right)^i}{1 - \left(\frac{q}{p}\right)^k} = \frac{1 - \left(\frac{1-p}{p}\right)^i}{1 - \left(\frac{1-p}{p}\right)^k}$$

Example: Gambler's ruin

Case 2: Fair Game ($p = \frac{1}{2}$, so $p = q$)

- Here, $\frac{q}{p} = 1$, so the sum simplifies to:

$$P_i = P_1 \sum_{j=0}^{i-1} 1 = i \cdot P_1$$

- Applying the boundary condition $P_k = 1$: $1 = k \cdot P_1 \Rightarrow P_1 = \frac{1}{k}$
- Thus:

$$P_i = \frac{i}{k}$$

- Final Answer:** The probability that gambler A wins is:

$$P_i = \begin{cases} \frac{1 - \left(\frac{1-p}{p}\right)^i}{1 - \left(\frac{1-p}{p}\right)^k} & \text{if } p \neq \frac{1}{2} \\ \frac{i}{k} & \text{if } p = \frac{1}{2} \end{cases}$$